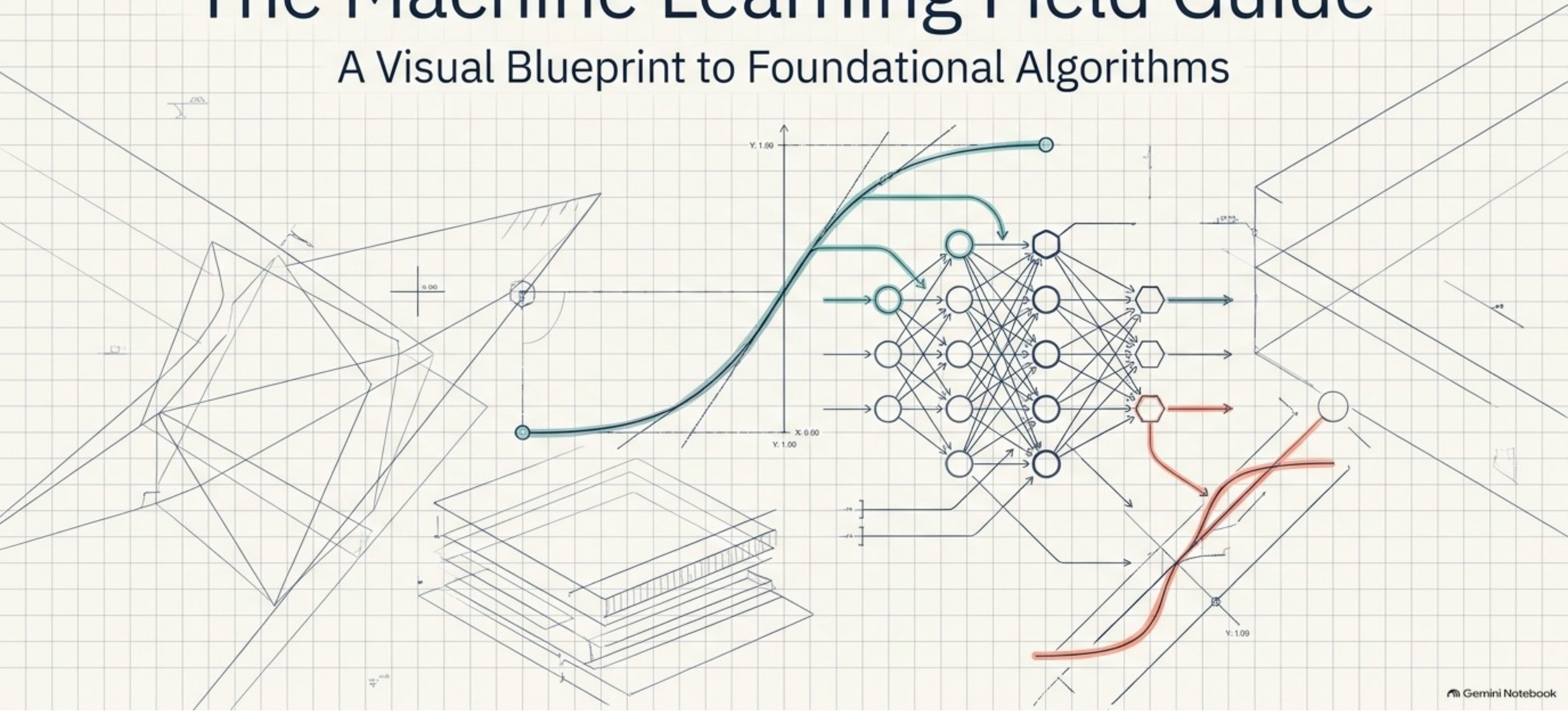
The Machine Learning Field Guide

A Visual Blueprint to Foundational Algorithms



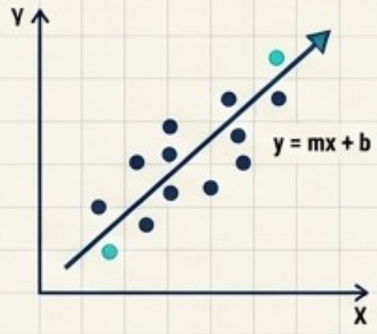
Gemini Notebook

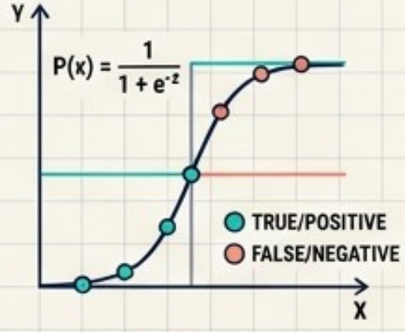
The Five Philosophies of Machine Learning

Category 1: The Baselines (Lines & Curves)

Algorithms: Linear & Logistic Regression

Mechanism: Fitting continuous shapes to data.

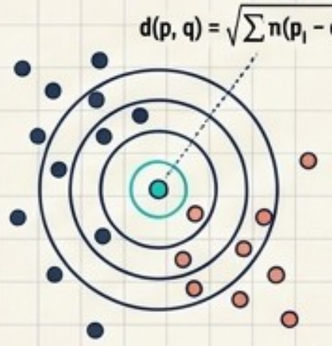




Category 2: The Observers (Distance & Probability)

Algorithms: K-Nearest Neighbors & Naive Bayes

Mechanism: Using raw existing data to map proximity or calculate odds.



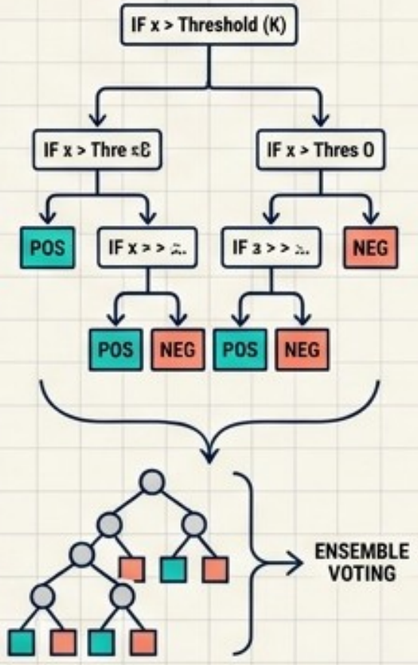
CLASS A
CLASS B

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Category 3: The Splitters (Logic)

Algorithms: Decision Trees & Random Forests

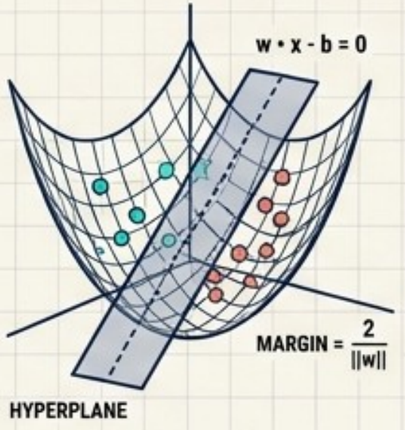
Mechanism: Segmenting data via sequential thresholds and voting.



Category 4: The Transformers (Geometry)

Algorithms: Support Vector Machines

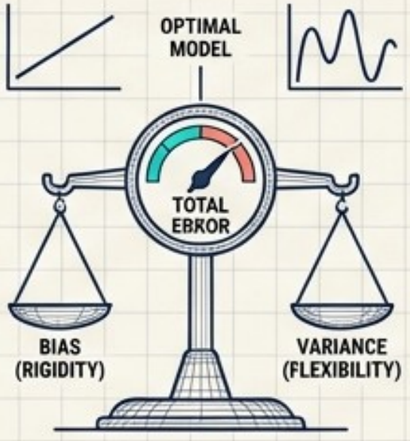
Mechanism: Warping dimensional space to find mathematical boundaries.



Category 5: The Synthesis (Evaluation)

Focus: The Bias-Variance Tradeoff

Mechanism: Balancing the rigidity and flexibility of models.



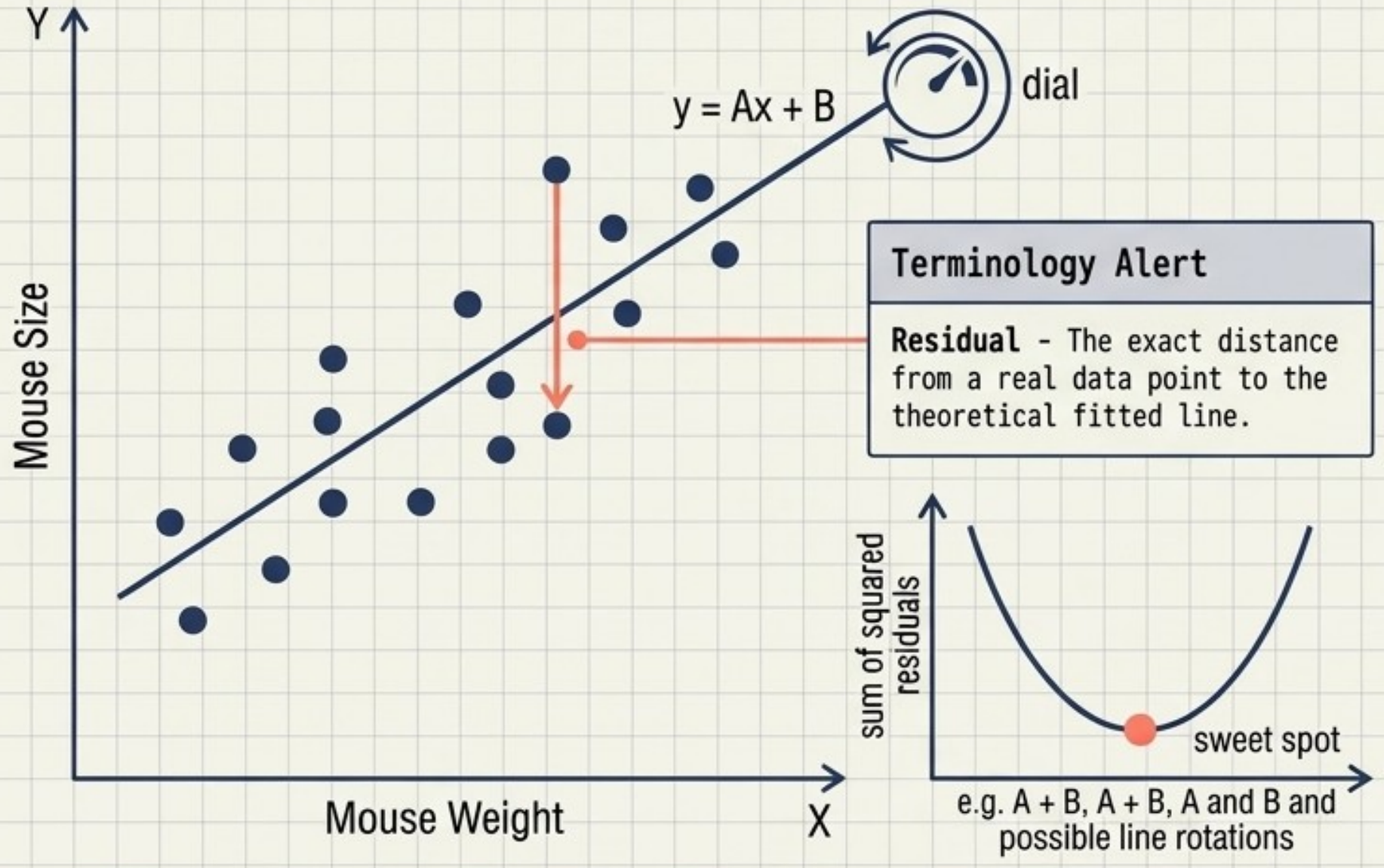
$$\text{Error} \approx \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

Gemini Notebook

Linear Regression

"Draw the line of best fit by minimizing the mistakes."

The Visual



Anatomy Layout

Input

Continuous numeric data (e.g., Weight).

Core Mechanism

Least Squares. The model systematically rotates a line (changing the slope A and intercept B), measures the distance from the line to every data point, squares that distance, and sums them. The rotation with the lowest sum wins.

Output

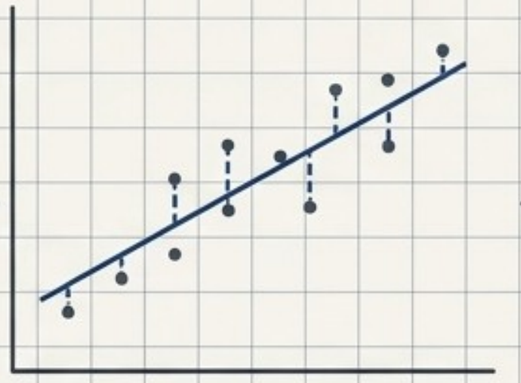
A predictive mathematical equation: $y = Ax + B$.

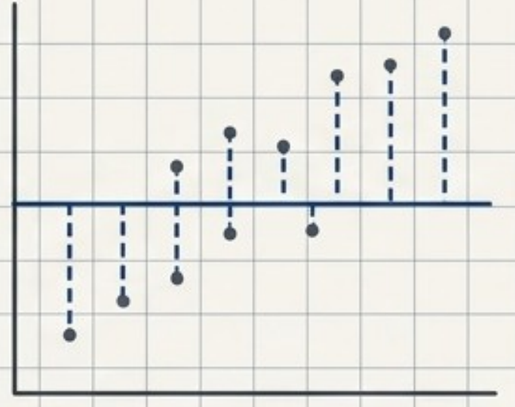
Gemini Notebook

Visualizing R-Squared: How Good is the Fit?

R-Squared is simply a percentage. It tells us how much of the original variation in the data is explained by our new line.

$$R^2 = \frac{\text{(Variance around the Mean - Variance around the Fit)}}{\text{Variance around the Mean}} = \text{Visual Fraction}$$





Insight

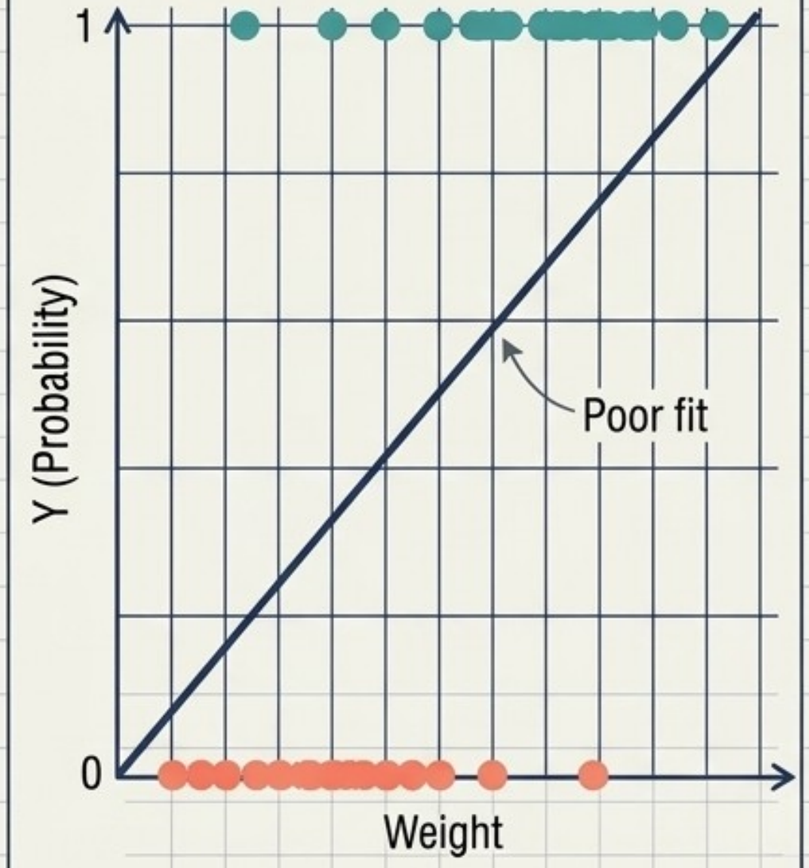
If R-Squared = 0.60, knowing a mouse's weight explains 60% of the variance in its size. The model isn't perfect, but it is 60% better than blindly guessing the average.

Gemini Notebook

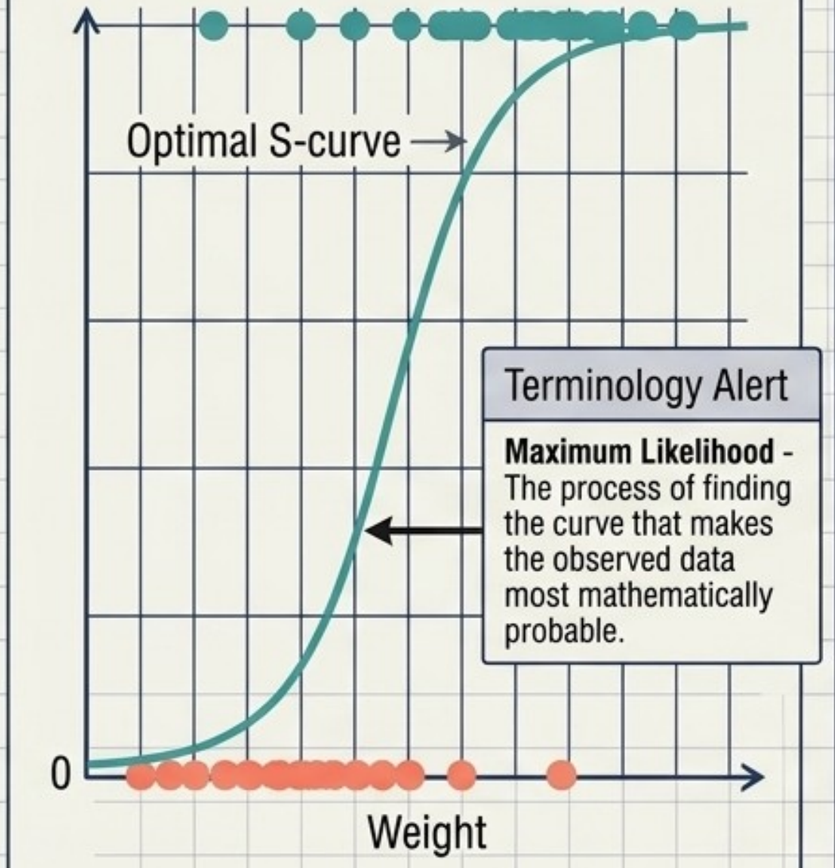
Logistic Regression

"Bend the line into a curve to predict true/false probabilities."

Linear Failure



Logistic Fit



Terminology Alert

Maximum Likelihood - The process of finding the curve that makes the observed data most mathematically probable.

Anatomy Layout

Input

Continuous or discrete data targeting a binary outcome (e.g., Obese: Yes/No).

Core Mechanism

Maximum Likelihood. Instead of measuring residuals (distance), the model shifts the S-curve until the combined mathematical likelihood of observing the actual data is maximized.

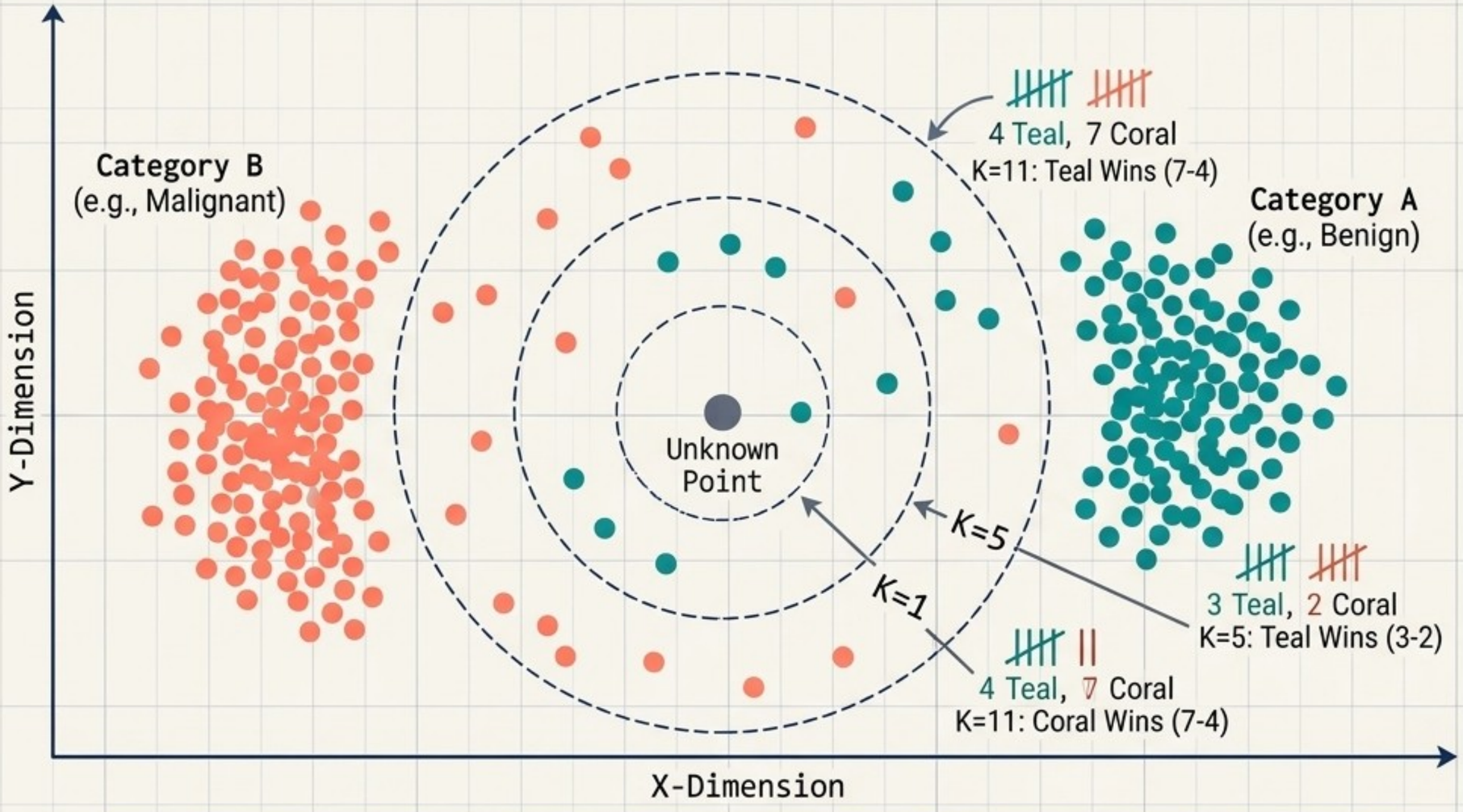
Output

A probability score between 0 and 1, allowing for a 50% classification threshold.

Gemini Notebook

K-Nearest Neighbors (KNN)

"Tell me who your neighbors are, and I'll tell you who you are."



Anatomy Layout

Input

New, unclassified data plotted in the exact same spatial dimensions as the known training data.

Core Mechanism

Distance Measurement & Majority Voting.

The algorithm identifies the K closest data points.

Output

A class prediction based strictly on which category gets the most votes from the neighbors.

Parameter Focus: K

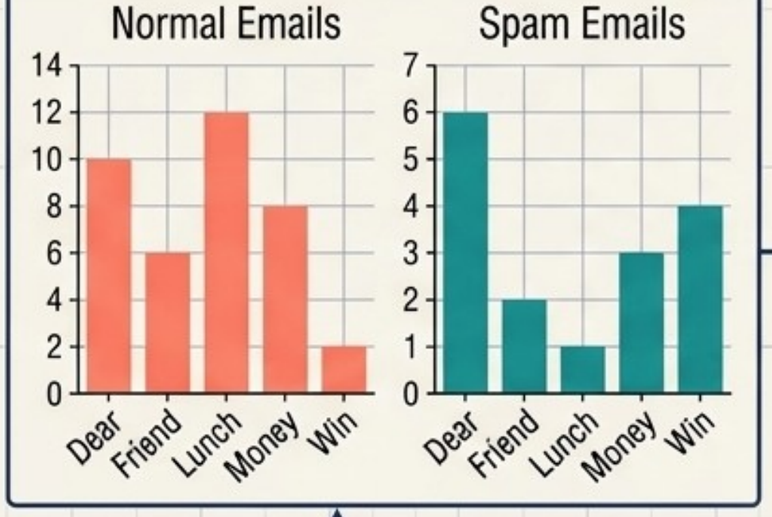
Low K (e.g., 1) is noisy and susceptible to outliers. High K smooths the decision but risks allowing massive categories to outvote small, legitimate ones.

Gemini Notebook

Naive Bayes

"Language is just a bag of words with independent probabilities."

Training Histogram



Incoming spam email with content: "Lunch Money Money."

Prediction Engine

Prior Probability (Spam Guess)		Probability of "Lunch"		Probability of "Money"^3		Proportional Probability Score
P(Spam) = 0.5	×	P('Lunch'|Spam) = 0.02	×	P('Money'|Spam)^3 = 0.15^3 = 0.003375	=	P(Spam | 'Lunch Money Money') ∝ 0.5 × 0.02 × 0.003375 = 0.00003375

Terminology Alert

Alpha - A pseudocount (usually 1) added to all features.

Why? If the word "Lunch" never appeared in training spam, its probability is 0. Multiplying anything by 0 destroys the calculation. Alpha ensures no probability is exactly zero.

The Naive Flaw

It ignores grammar and word order (high bias), treating "Dear Friend" the exact same as "Friend Dear"."

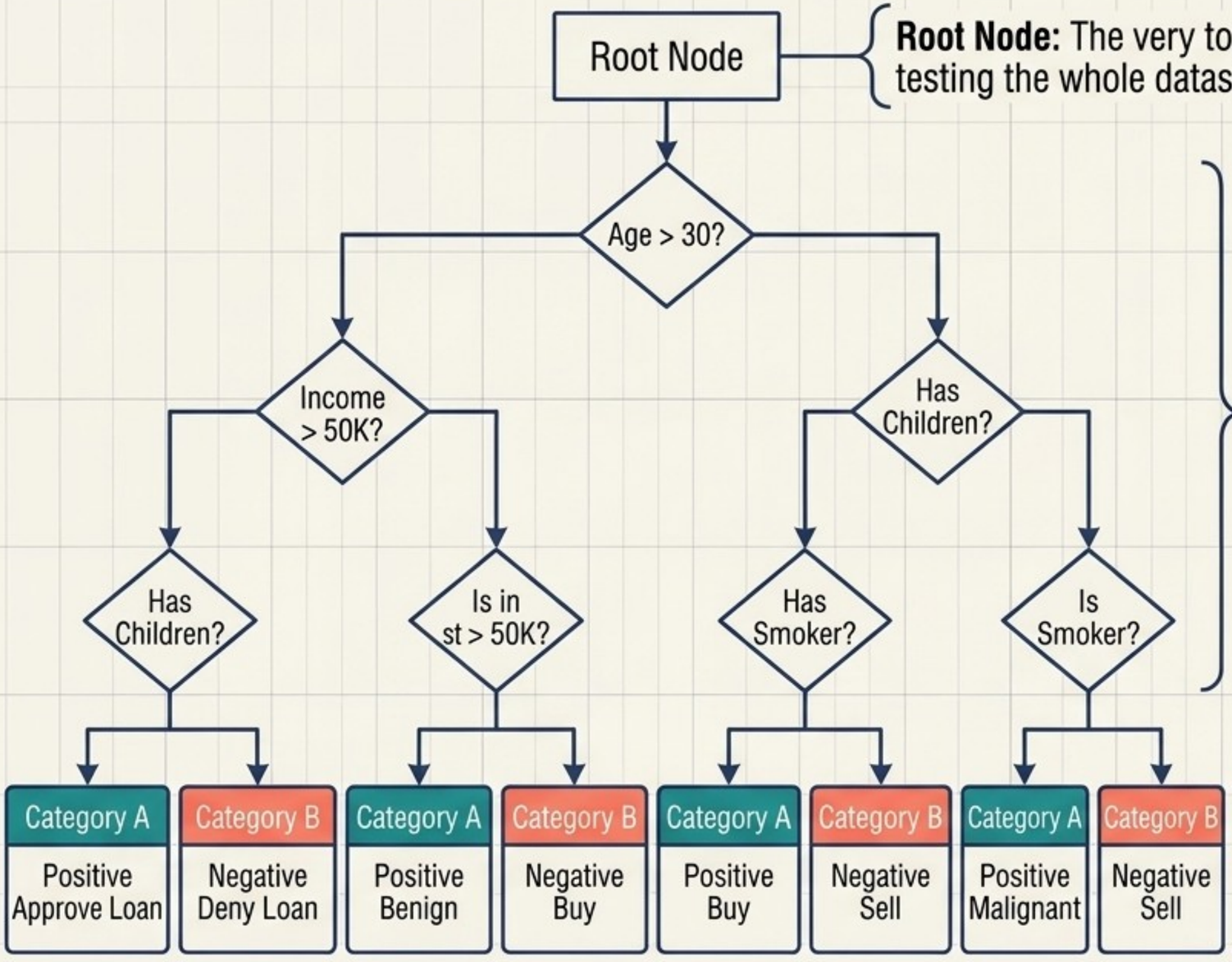
Anatomy Layout

Input	Core Mechanism	Output
Discrete features (e.g., word counts in Normal vs. Spam emails).	Multiplying the Prior Probability (initial guess) by the individual likelihoods of each observed feature.	A proportional probability score used to classify the text.

Gemini Notebook

Decision Trees

"Ask a sequence of yes/no questions to divide and conquer the data."



Anatomy Layout

Input

Mixed data types (numeric ages, categorical yes/no preferences).

Core Mechanism

Sequential logic thresholds testing all variables to find the most mathematically distinct split.

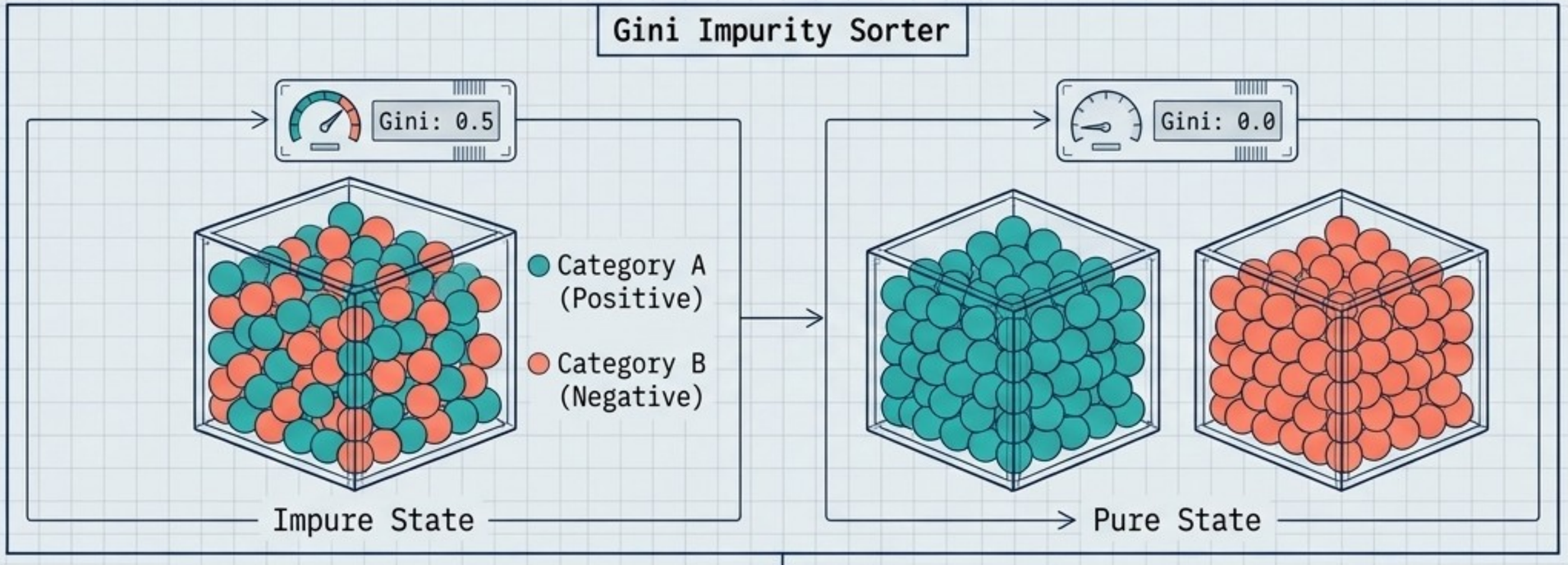
Output

A definitive classification based on the majority inside the final grouping.

Gemini Notebook

The Math of the Split: Gini Impurity

Trees evaluate splits by measuring the 'impurity' in the resulting leaves. The goal is a score of exactly 0.



The Visual Formula: 1 - P(yes)^2 - P(no)^2

(Starts at 1, subtracts the squared probability of every possible outcome).

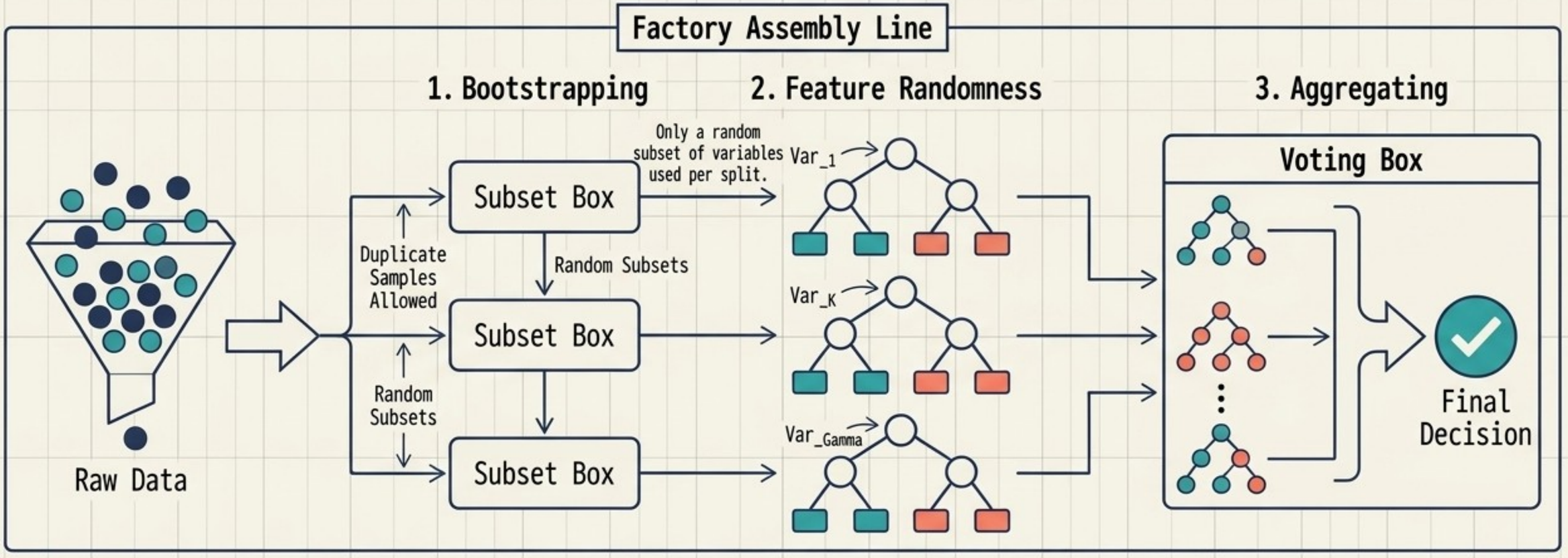
Mechanism

The algorithm calculates the weighted average of the Gini impurity for every possible threshold (e.g., Age < 9.5 vs. Age < 15) and permanently chooses the split with the lowest score.

Gemini Notebook

Random Forests

"Crowdsource the decision to a diverse, randomized committee to prevent memorization."



Core Mechanism: Bagging (Bootstrapping + Aggregating).

Bootstrapping creates random sub-datasets of the same size, allowing duplicate samples. **Feature Randomness** forces trees to consider only a random subset of variables at each split, ensuring diverse, uncorrelated trees.

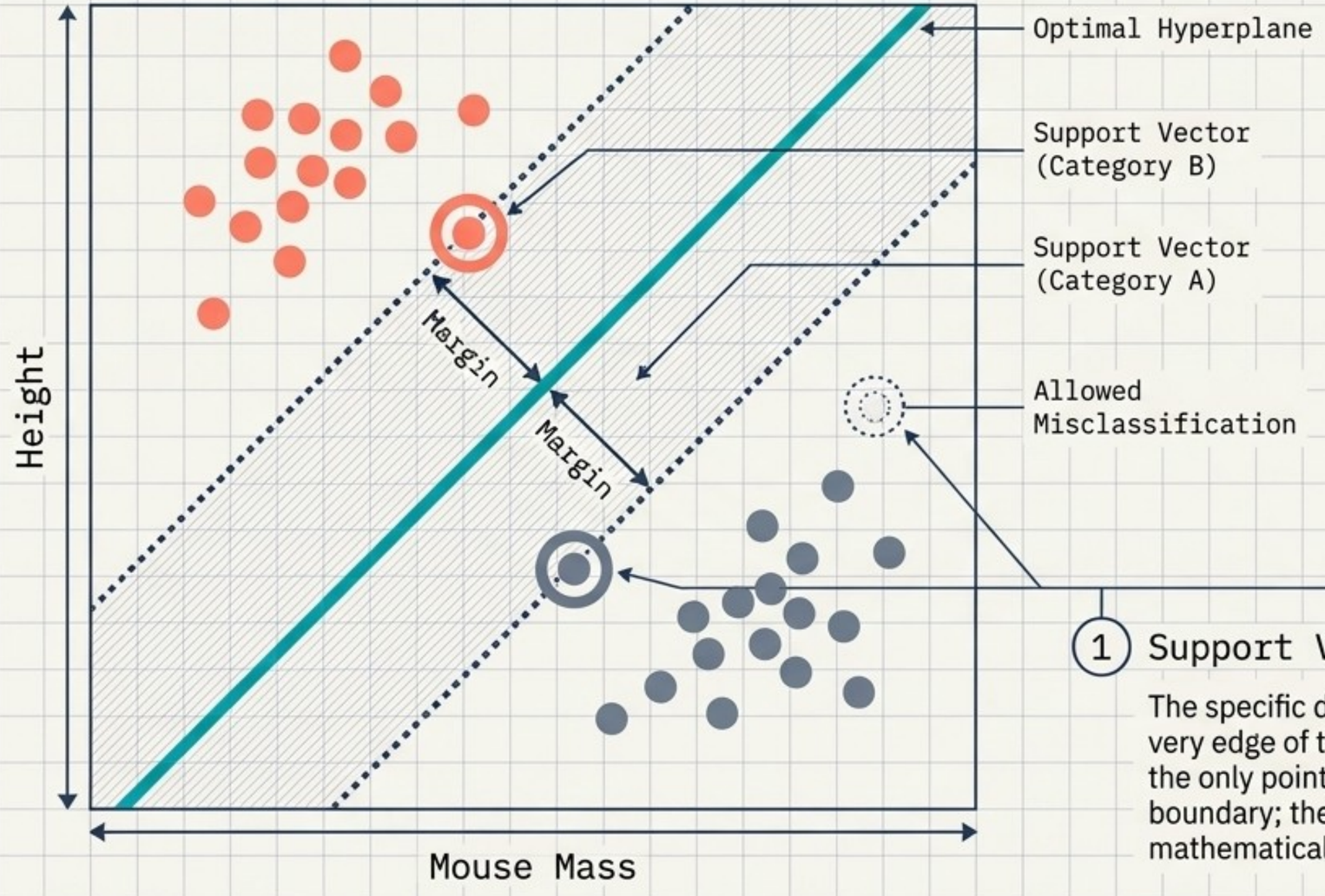
Terminology Alert: Out-of-Bag Error.

Because bootstrapping leaves roughly 1/3 of the original data out of each tree, this leftover data is immediately used to test the tree's accuracy without needing a separate test dataset.

Gemini Notebook

Support Vector Machines (SVM)

Philosophy: "Draw the widest possible street between categories."



Input

Multi-dimensional spatial data.

Core Mechanism

Finding a hyperplane (a line in 2D, a plane in 3D) that maximizes the margin of empty space between classes.

Output

A rigid geometric boundary.

1 Support Vectors

The specific data points on the very edge of the margin. They are the only points that dictate the boundary; the rest of the data is mathematically ignored.

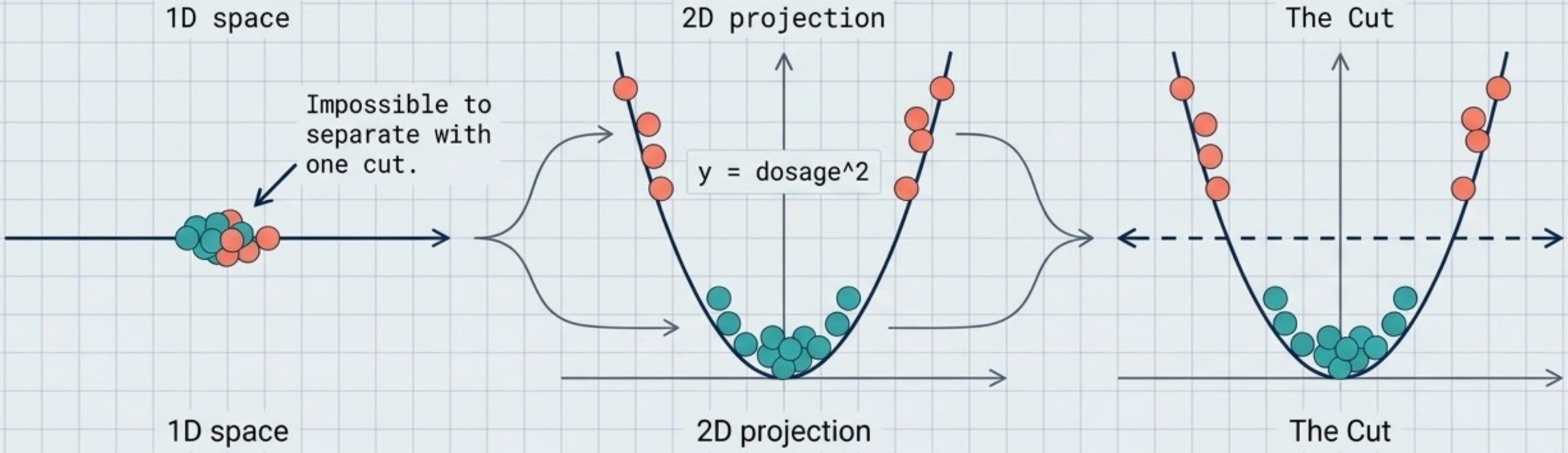
2 Soft Margin

Allowing deliberate misclassifications inside the margin to prevent the model from overreacting to single outliers.

Gemini Notebook

The Kernel Trick

"When a flat line won't cut it, fold the dimensional space."



Core Concept

The Polynomial Kernel calculates the dot products (relationships) of data as if it were plotted in higher dimensions (like squaring the dosage).

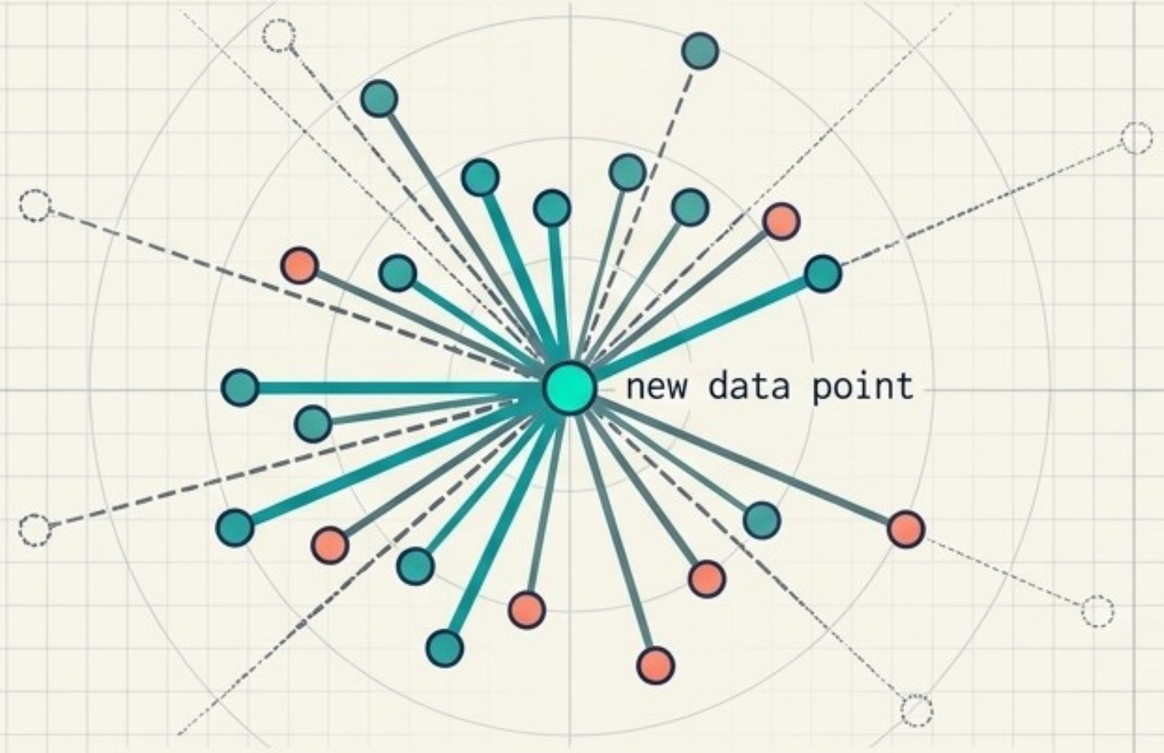
The "Trick"

It computes these high-dimensional mathematical relationships without the massive computational power required to actually transform and plot the raw data.

Gemini Notebook

The Infinite Dimension: RBF Kernel

"Calculate relationships across infinite dimensions."



Taylor Series expansion fading out into infinity:

$$
\begin{aligned}
K(\mathbf{x}, \mathbf{x}') &= \exp(-\gamma \left\| \mathbf{x} - \mathbf{x}' \right\|^2) \\
&= 1 - \gamma \left\| \mathbf{x} - \mathbf{x}' \right\|^2 \\
&\quad + \frac{\gamma^2}{2!} \left\| \mathbf{x} - \mathbf{x}' \right\|^4 \\
&\quad - \frac{\gamma^3}{3!} \left\| \mathbf{x} - \mathbf{x}' \right\|^6 + \cdots + \infty
\end{aligned}
$$

Core Concept

The Radial Kernel relies on the mathematical properties of a Taylor Series expansion (which splits a function into an infinite sum). This behaves practically like a weighted K-Nearest Neighbors model.

Parameter Focus: Gamma

Gamma scales the squared distance between points.

- High Gamma = Only the immediate, closest neighbors influence the decision.
- Low Gamma = Influence reaches far across the dataset, smoothing the boundaries.

Gemini Notebook

The Algorithm Blueprint: Master Matrix

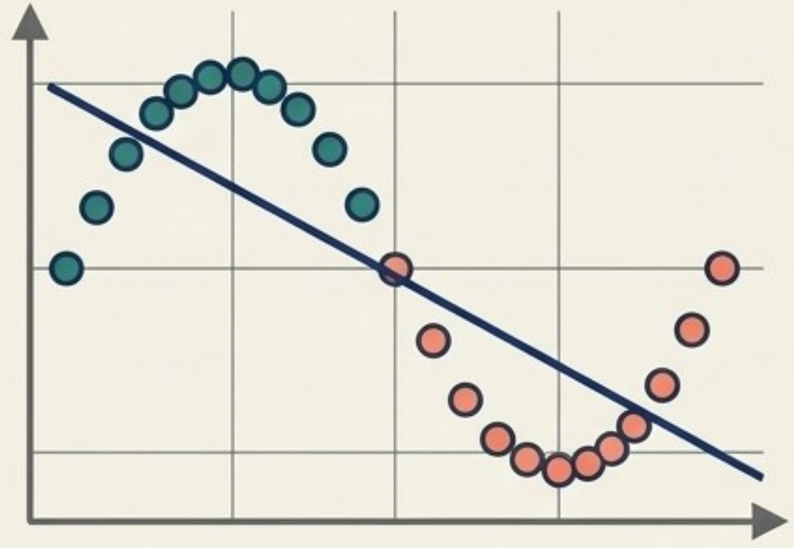
Algorithm	Primary Philosophy	Key Parameters to Tune	Sensitivity to Outliers
Linear Reg	Minimize Squares	None	High
Logistic Reg	Maximize Likelihood	None	High
KNN	Spatial Voting	K	Medium
Naive Bayes	Independent Probabilities	Alpha	Low
Decision Trees	Logical Splitting	Depth / Limits	High
Random Forest	Ensembles	Trees / Variables per split	Low
SVM	Maximize Margins	D / Gamma / Soft Margin	Low

Gemini Notebook

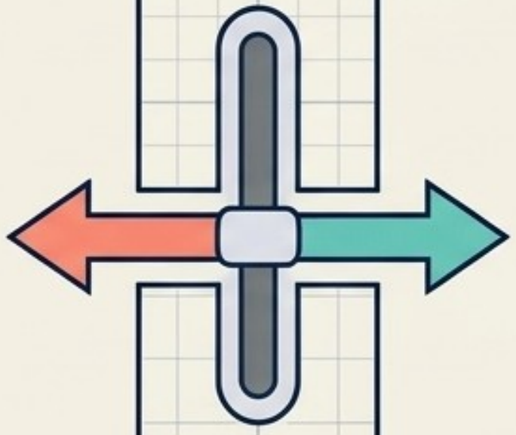
The Universal Tension: Bias vs. Variance

"Every parameter exists to solve one problem: tuning the perfect balance between rigidity and flexibility."

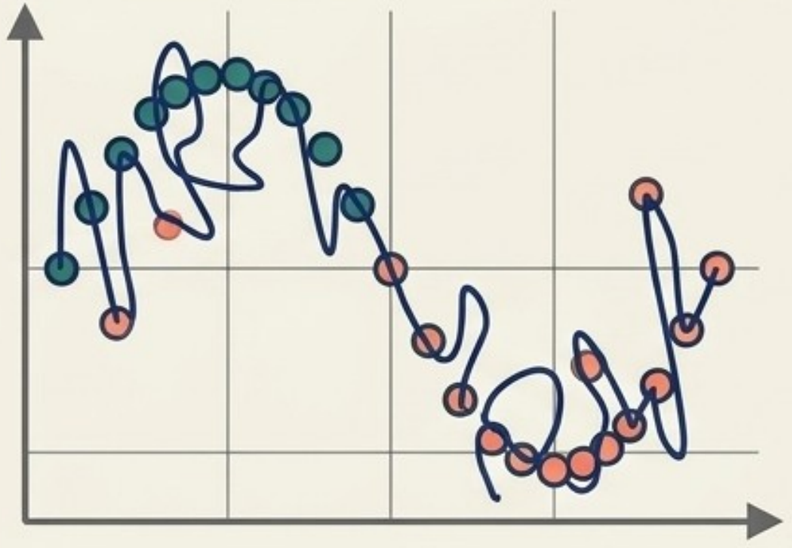
Underfitting (Oversimplifying). Rigid rules, high assumptions.



Models: Naive Bayes, Low K KNN, Hard Margin SVM.



Overfitting (Memorizing). Infinitely flexible, highly sensitive to noise.



Models: Deep Decision Trees, High Gamma RBF.

The Blueprint's Goal:
Machine learning is not about finding the perfect mathematical truth; it is about turning the dials to navigate this exact trade-off.

Gemini Notebook